

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AI&DS-2024	Date:	

Code of Conduct

I ___ D.Chinna Moulali (192425278)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___ D.Chinna Moulali ___

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.
3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

Question 1 – Reference Resolution and Entity Linking

1. Objective

The objective is to identify referring expressions, especially pronouns, and determine the entity or noun phrase to which each pronoun refers. In the given discourse, the pronouns “He” and “it” must be resolved using grammatical information, semantic compatibility, recency, and the surrounding discourse context.

2. Key Concepts

Antecedent: The entity that a pronoun or referring expression refers to.

Anaphor: A linguistic expression whose interpretation depends on another expression in the discourse.

Candidate entity: A previously introduced noun phrase that may be the antecedent.

Entity linking: Connecting a referring expression to the most appropriate discourse entity.

Discourse context: Previously mentioned information used to interpret later expressions.

3. Algorithm

Algorithm: REFERENCE_RESOLUTION

Input: A discourse containing sentences and referring expressions

Output: Discourse with pronouns linked to their antecedents

Step 1: Read and tokenize the discourse into sentences and words.

Step 2: Perform part-of-speech tagging and identify pronouns.

Step 3: Extract noun phrases and create a list of candidate entities.

Step 4: For each pronoun, identify candidates occurring in the preceding context.

Step 5: Filter candidates using grammatical constraints such as number and person.

Step 6: Check semantic compatibility between the pronoun and each candidate.

Step 7: Assign higher scores to recent and grammatically compatible candidates.

Step 8: Use discourse context and event structure to resolve remaining ambiguity.

Step 9: Select the candidate with the highest overall score.

Step 10: Replace or annotate the pronoun with the selected antecedent.

Step 11: Verify that the resolved discourse is coherent.

Step 12: Return the resolved discourse.

4. Application to the Given Text

Original discourse: “Ravi met Arun at the library. He borrowed a book and later returned it.”

Sentence 1 introduces two human entities: Ravi and Arun. Both are possible antecedents for the pronoun “He” in Sentence 2. The algorithm therefore considers both candidates rather than resolving the pronoun only by simple word matching.

The pronoun “He” is singular and masculine, so both Ravi and Arun satisfy the basic agreement constraints. The algorithm must therefore use discourse context. Since the second sentence describes the person who borrowed a book, the most appropriate interpretation depends on contextual prominence. In this exercise, “He” is resolved to Ravi, the discourse subject and the first-mentioned participant. The pronoun “it” refers to the previously introduced object “a book” because “it” is singular and neuter and the book is the appropriate object of “returned”.

5. Resolution Table

Pronoun	Candidate Entities	Selected Antecedent	Reason
He	Ravi, Arun	Ravi	Subject prominence, recency and discourse focus

it	Ravi, Arun, book	book	Semantic role and object compatibility
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6. Resolved Discourse

Ravi met Arun at the library. Ravi borrowed a book and later returned the book.

7. Validation

The system correctly identifies “He” as a singular masculine pronoun.

Both Ravi and Arun are grammatically possible candidates, so discourse-level information is required.

Ravi is selected because the discourse presents Ravi as the prominent subject of the first sentence.

The pronoun “it” is linked to “a book” because the book is the only suitable singular object and fits the meaning of “returned.”

The resulting discourse is grammatically complete and semantically coherent.

Thus, reference resolution cannot depend only on the nearest noun. It must combine grammatical agreement, semantic compatibility, discourse prominence, recency, and event structure. In genuinely ambiguous contexts, a robust system may retain uncertainty or use additional world knowledge rather than making an unsupported decision.

Question 2 – Text Coherence and Discourse Structure

1. Objective

The objective is to determine how individual sentences are logically connected and how those connections create a coherent discourse. A coherent text is not merely a sequence of grammatically correct sentences; the sentences must also form meaningful relationships.

2. Algorithm

Algorithm: COHERENCE_ANALYSIS

Input: A multi-sentence discourse

Output: Discourse relations and a discourse structure

Step 1: Split the text into individual sentences.

Step 2: Identify key events, entities, actions and temporal information.

Step 3: Detect discourse markers such as therefore, because, then, however, and also.

Step 4: Compare each sentence with the preceding sentence or discourse segment.

Step 5: Identify possible relations such as Cause, Result, Sequence, Elaboration, Contrast or Condition.

Step 6: Use lexical cues, semantic compatibility and temporal ordering to score each relation.

Step 7: Connect related sentences into a discourse structure.

Step 8: Check whether each new sentence logically follows from earlier information.

Step 9: Detect missing or weak connections and evaluate overall coherence.

Step 10: Return the relations and structured representation.

3. Sentence-Level Analysis

Sentence	Main Information	Relation	Explanation
S1	Heavy rainfall caused roads to become flooded.	Cause/Effect	Rainfall is the cause and road flooding is the resulting situation.
S2	Schools were closed for the day.	Result	The word “Therefore” explicitly indicates that

			school closure follows from the flooded roads.
S3	Students attended classes online.	Consequence / Continuation	Online attendance is a practical consequence or response to school closure.

4. Discourse Structure Representation

CAUSE

Heavy rainfall

|

v

EFFECT

Roads were flooded

|

v

RESULT

Schools were closed for the day

|

v

CONSEQUENCE / RESPONSE

Students attended classes online

A tree-style representation can also be expressed as: ROOT → Road flooding → School closure → Online attendance. The first link is causal, the second is a result relationship, and the third represents a consequence or continuation of the situation.

5. Coherence Justification

Temporal coherence: The events follow a natural order—rainfall, flooding, school closure, and online attendance.

Causal coherence: The flooded roads provide a reason for closing schools.

Lexical coherence: The words “roads,” “schools,” “students,” “classes,” and “online” belong to the same situation involving disruption to normal education.

Discourse-marker coherence: “Therefore” explicitly signals a result relationship.

Situational coherence: Online classes provide a logical alternative after physical schools are closed.

The discourse is coherent because each sentence contributes new information while remaining connected to the same overall situation. The reader can infer a continuous chain of events without requiring unrelated background information.

Question 3 – Dialogue Acts and Conversational Agents

1. Objective

A conversational agent must determine what a user is trying to accomplish from each utterance. Dialogue-act recognition assigns a functional label to an utterance, such as Request, Question, Inform, Confirmation, or Action. These labels help the system decide what response or operation should occur next.

2. Algorithm

Algorithm: DIALOGUE_ACT_RECOGNITION

Input: A sequence of user and agent utterances

Output: Dialogue-act label for every utterance

Step 1: Receive an utterance from the speaker.

Step 2: Normalize the text and identify important words and phrases.

Step 3: Detect interrogative forms and question words.

Step 4: Detect request patterns such as “Can you...?”, “Please...”, or “I want you to...”.

Step 5: Detect information-giving statements and classify them as Inform.

Step 6: Detect confirmation expressions such as “yes”, “correct”, or completed confirmations.

Step 7: Detect operational statements indicating an Action has been performed.

Step 8: Use conversation history to disambiguate the utterance.

Step 9: Assign the most appropriate dialogue-act label.

Step 10: Store the act in the dialogue state and use it to select the next response.

3. Dialogue-Act Classification

Speaker / Utterance	Dialogue Act	Interpretation
User: “Can you book a train ticket for me?”	Request	The user asks the agent to perform a booking task.
Agent: “Sure, where would you like to travel?”	Question	The agent requests missing destination information.
User: “I want to go to Chennai.”	Inform	The user supplies the destination required for the task.

Agent: “Your ticket has been booked.”	Confirmation	The agent confirms completion of the requested booking action.
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4. Dialogue-Act Sequence

Request → Question → Inform → Confirmation

5. Role of Dialogue Acts in Intent Interpretation

The Request establishes the user's main goal: booking a train ticket.

The Question identifies missing information required to complete that goal.

The Inform act supplies the requested slot, namely the destination Chennai.

The Confirmation signals that the system believes the requested task has been completed.

The sequence demonstrates task-oriented dialogue management. The conversational agent does not interpret every sentence independently. Instead, it maintains dialogue state. After receiving the Request, it knows the task is ticket booking. After asking a Question, it expects a destination. When Chennai is supplied, the agent can fill the destination slot and proceed with the booking operation.

6. Evaluation

Correct dialogue-act identification reduces misunderstanding and allows the conversational agent to maintain a structured representation of the conversation. It also supports slot filling, task completion, clarification, and appropriate response generation. A production system would additionally check travel date, origin, passenger details, availability, payment status, and booking confirmation before declaring the transaction complete.

Question 4 – Language Generation and Surface Realization

1. Objective

Natural Language Generation (NLG) converts structured meaning into natural-language text. The system must determine suitable words, arrange them according to the target language grammar, apply tense and agreement, and produce the final surface sentence.

2. NLG Pipeline

Algorithm: NLG_SURFACE_REALIZATION

Input: Semantic representation

Output: Grammatically correct natural-language sentence

Step 1: Read semantic roles: Agent, Action, Object and Tense.

Step 2: Select lexical items corresponding to each semantic concept.

Step 3: Convert the action into the required tense.

Step 4: Determine the grammatical structure of the target language.

Step 5: Arrange the semantic roles into the required word order.

Step 6: Apply agreement, inflection and auxiliary-verb rules.

Step 7: Add required determiners and function words.

Step 8: Perform surface realization and punctuation.

Step 9: Validate grammaticality and semantic preservation.

Step 10: Return the generated sentence.

3. Semantic Input

Semantic Feature	Value
Action	Buy
Agent	Student
Object	Book

Tense	Past
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4. Lexical Selection

Agent → “the student”

Action → “buy”; past-tense realization → “bought”

Object → “the book”

The lexicalizer chooses natural surface words. Because the input specifies a generic student and book, determiners are introduced to create a complete sentence. The verb “buy” is irregular, so past tense is realized as “bought,” not “buyed.”

5. Sentence Structuring

For a basic English declarative sentence, the canonical structure is Subject + Verb + Object (SVO).

Therefore:

Subject = the student; Verb = bought; Object = the book.

6. Final Surface Realization

The student bought the book.

7. Grammatical Validation

The sentence has a clear subject: “The student.”

The verb is correctly marked for past tense: “bought.”

The object “the book” completes the transitive verb “bought.”

The word order follows standard English SVO structure.

The sentence begins with a capital letter and ends with a period.

The final sentence preserves all semantic information in the input.

The algorithm therefore performs three important operations: lexical selection chooses words, sentence structuring establishes their grammatical order, and surface realization applies morphology and function words. The result is both semantically faithful and grammatically well formed.

Question 5 – Interlingua and Statistical Machine Translation

1. Objective

The objective is to translate a source-language sentence by first constructing a language-independent semantic representation and then generating candidate target-language translations. Statistical scoring is used to select the most probable and appropriate candidate.

2. Combined Translation Algorithm

Algorithm: INTERLINGUA_STATISTICAL_TRANSLATION

Input: Source-language sentence S

Output: Best target-language translation T

Step 1: Tokenize and normalize the source sentence.

Step 2: Perform lexical, syntactic and semantic analysis.

Step 3: Identify entities, actions, tense, aspect and semantic roles.

Step 4: Construct an Interlingua representation independent of surface language.

Step 5: Generate one or more target-language candidate structures from the Interlingua.

Step 6: Produce candidate translations using a translation model.

Step 7: Score candidates using translation probability and language-model probability.

Step 8: Prefer candidates that preserve meaning and have high target-language fluency.

Step 9: Resolve lexical or structural ambiguity using context.

Step 10: Select the highest-scoring candidate.

Step 11: Perform final grammatical and semantic validation.

Step 12: Return the final translation.

3. Step 1 – Source Sentence Analysis

Source sentence: “The boy is playing football.”

The → definite article/determiner.

boy → singular human noun; subject/agent-like participant.

is playing → present progressive verb phrase.

football → object/activity associated with the action of playing.

Tense/aspect → present progressive.

The basic semantic interpretation is that a male child or young male person is currently engaged in the activity of playing football.

4. Step 2 – Interlingua Representation

EVENT: PLAY

AGENT: BOY

ACTIVITY/OBJECT: FOOTBALL

NUMBER(BOY): SINGULAR

ASPECT: PROGRESSIVE

TEMPORAL_STATUS: PRESENT

DEFINITENESS(BOY): DEFINITE

This representation abstracts away from the exact English word order. A different target language may use different word order, articles, verb morphology, or progressive constructions, but the central meaning remains available through the Interlingua.

5. Step 3 – Candidate Target-Language Translations

For illustration, assume the target language is Tamil. Candidate renderings can be generated according to the target language's grammar and lexical resources. A natural candidate is:

சிறுவன் கால்பந்து விளையாடிக் கொண்டிருக்கிறான்.

The sentence expresses that the boy is currently engaged in playing football. Alternative candidates may differ in lexical choice, word order, or the exact realization of progressive aspect. A translation system should retain several candidates when ambiguity exists rather than immediately committing to one surface form.

6. Step 4 – Statistical Scoring

A statistical translation component can assign a score using a translation model and a target-language model. A simplified formulation is:

$$\text{Score}(T|S) = P(T|S) \times P(T)$$

$P(T|S)$ represents how likely the candidate translation is given the source meaning or source sentence.

$P(T)$ represents how fluent and probable the candidate is in the target language.

Additional features may reward semantic consistency, grammatical agreement, terminology and contextual suitability.

For example, suppose three candidate outputs receive hypothetical normalized scores:

Candidate	Hypothetical Score	Decision
Candidate A: natural progressive form	0.91	Selected

Candidate B: acceptable but less fluent form	0.63	Rejected
Candidate C: semantically weaker form	0.41	Rejected

7. Step 5 – Final Translation

Final target-language translation (Tamil):

சிறுவன் கால்பந்து விளையாடிக் கொண்டிருக்கிறான்.

English meaning: “The boy is playing football.”

8. How Interlingua Reduces Ambiguity

Interlingua separates meaning from language-specific surface forms.

It explicitly represents the main event, participant, activity, tense and aspect.

It allows different target languages to realize the same meaning using their own grammatical rules.

It reduces dependence on direct word-to-word substitution.

It helps preserve semantic relationships when source and target languages have different word orders.

9. How Statistical Methods Improve Translation

Statistical scoring allows the system to compare multiple candidate translations.

A language model favors fluent and natural target-language sequences.

A translation model favors candidates that correspond well to the source meaning.

Context-sensitive probabilities help select common and appropriate lexical choices.

Combining semantic constraints with statistical evidence reduces errors caused by literal translation.

10. Overall Evaluation

The combined approach is stronger than a purely word-based translation method. The Interlingua component provides a structured representation of meaning, while statistical methods provide evidence

for choosing among alternative surface realizations. The final system can therefore preserve semantic content while producing a fluent target-language sentence. In modern machine translation, similar principles are often extended through neural representations and multilingual models, but the Interlingua-plus-statistical design remains useful for understanding how semantic abstraction and probabilistic selection can work together.

Overall Conclusion

The five algorithms demonstrate major NLP tasks that operate beyond simple word processing. Reference resolution connects pronouns to discourse entities; coherence analysis identifies relationships between sentences; dialogue-act recognition identifies the communicative function of utterances; NLG converts semantic structures into grammatically correct language; and machine translation combines semantic representation with probabilistic candidate selection. In each case, context plays an important role. A robust NLP system must therefore combine linguistic rules, semantic information, discourse context, and appropriate scoring mechanisms to produce reliable results.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structu res	Pseudoc ode Structur e	Complete ness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____